R13

Code No: 115EH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech III Year I Semester Examinations, November/December - 2018
OPERATING SYSTEMS
(Common to CSE, IT)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a)	What are the goals of operating system?	[2]
b)	What is an Operating system? What are the various OS Components?	[3]
c)	What is critical section problem?	[2]
d)	What is preemptive Scheduling? How is it different from non-preemptive scheduling?	[3]
e)	What is Virtual Memory? Why is it required?	[2]
f)	Compare LRU and Optimal Replacement Algorithms.	[3]
g)	What are the files attributes up dated during file creation?	[2]
h)	What are the methods for accessing the file?	[3]
i)	Define deadlock.	[2]
j)	Write down the principles of protection.	[3]

PART - B

(50 Marks)

2. Describe evolution of operating system in detail. [10]

OR

3. What is the need for system calls? Explain the types of system calls provided by an operating system with respect to memory management. [10]

4. Explain FCFS, RR and SJF scheduling algorithm with illustrations. [10]

OR

5. Explain about multiple-processor scheduling and real time scheduling. [10]

6.a) What is the need of demand paging? Explain briefly.
b) Explain in detail about segmentation on with paging technique. [5+5]

OR

7.a) Explain in detail about paging technique.
b) Explain in detail Contiguous Memory Allocation. [5+5]

8. Explain in detail about file system structure and implementation. [10]

OR

9. Explain in detail about free space management. [10]

10. How can deadlock be detected and recovered? Explain in detail with relevant example. [10]

OR

11.a) What is access matrix? What are various methods to implement it?
b) Explain Capability-Based Protection system. [5+5]

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